

Mat 1575 HW1 : § 4.10

For the following exercises, show that $F(x)$ are antiderivatives of $f(x)$.

465. $F(x) = 5x^3 + 2x^2 + 3x + 1, f(x) = 15x^2 + 4x + 3$

Sol: check if $F'(x) = f(x)$: $F'(x) = 15x^2 + 4x + 3 = f(x) \checkmark$

Yes. $F(x)$ is an antiderivative of $f(x)$

For the following exercises, find the general form for the antiderivative of the function.

470. $f(x) = \frac{1}{x^2} + x$

471. $f(x) = e^x - 3x^2 + \sin x$

Let $F(x)$ be an antiderivative of $f(x)$

Sol. $f(x) = x^{-2} + x$

$$\begin{aligned} \Rightarrow F(x) &= \frac{x^{-2+1}}{-1} + \frac{x^{1+1}}{2} + C \\ &= -x^{-1} + \frac{x^2}{2} + C \\ &= -\frac{1}{x} + \frac{x^2}{2} + C \end{aligned}$$

Sol: $f(x) = e^x - 3x^2 + \sin(x)$

$$\begin{aligned} \Rightarrow F(x) &= e^x - \frac{3x^{2+1}}{3} + (-\cos(x)) + C \\ &= e^x - x^3 - \cos(x) + C \end{aligned}$$

For the following exercises, find the general form for the antiderivative $F(x)$ of each function $f(x)$.

476. $f(x) = \frac{1}{\sqrt{x}}$

Sol: $f(x) = \frac{1}{x^{\frac{1}{2}}} = x^{-\frac{1}{2}} \Rightarrow F(x) = \frac{x^{-\frac{1}{2}+1}}{\frac{1}{2}} + C = 2x^{\frac{1}{2}} + C$

477. $f(x) = (\sqrt{x})^3$

Sol: $f(x) = (x^{\frac{1}{2}})^3 = x^{\frac{3}{2}} \Rightarrow F(x) = \frac{x^{\frac{3}{2}+1}}{\frac{5}{2}} + C = \frac{2}{5}x^{\frac{5}{2}} + C$

481. $f(x) = \sec^2(x) + 1$ Sol: $F(x) = \tan(x) + x + C$

484. $f(x) = 0$ Sol: $F(x) = C$

For the following exercises, evaluate the integral.

490. $\int (-1) dx = -x + C$

492. $\int (4x + \sqrt{x}) dx = \int (4x + x^{\frac{1}{2}}) dx = \frac{4x^2}{2} + \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + C = 2x^2 + \frac{2}{3}x^{\frac{3}{2}} + C$

493. $\int \frac{3x^2 + 2}{x^2} dx = \int \left(\frac{3x^2}{x^2} + \frac{2}{x^2} \right) dx = \int (3 + 2x^{-2}) dx = 3x + \frac{2x^{-1}}{-1} + C = 3x - 2x^{-1} + C = 3x - \frac{2}{x} + C$

495. $\int (4\sqrt{x} + \sqrt[4]{x}) dx = \int (4x^{\frac{1}{2}} + x^{\frac{1}{4}}) dx = \frac{4x^{\frac{3}{2}}}{\frac{3}{2}} + \frac{x^{\frac{5}{4}}}{\frac{5}{4}} + C = \frac{8}{3}x^{\frac{3}{2}} + \frac{4}{5}x^{\frac{5}{4}} + C$

496. $\int (x^{-1/3} - x^{2/3}) dx = \frac{x^{\frac{2}{3}}}{\frac{2}{3}} - \frac{x^{\frac{5}{3}}}{\frac{5}{3}} = \frac{3}{2}x^{\frac{2}{3}} - \frac{3}{5}x^{\frac{5}{3}} + C$

For the following exercises, solve the initial value problem.

499. $f'(x) = x^{-3}$, $f(1) = 1$

Sol $f(x) = \frac{x^{-2}}{-2} + C$. Since $f(1) = 1$, then $1 = f(1) = \frac{1^{-2}}{-2} + C = -\frac{1}{2} + C$
 $\Rightarrow C = \frac{3}{2}$

$$\Rightarrow f(x) = -\frac{x^{-2}}{2} + \frac{3}{2}$$

500. $f'(x) = \sqrt{x} + x^2$, $f(0) = 2$ $f'(x) = x^{\frac{1}{2}} + x^2$

Sol. $f(x) = \frac{2}{3}x^{\frac{3}{2}} + \frac{x^3}{3} + C$. Since $f(0) = 2$, then $2 = f(0) = \frac{2}{3} \cdot 0 + \frac{0}{3} + C = C$
 $\Rightarrow C = 2$

$$\Rightarrow f(x) = \frac{2}{3}x^{\frac{3}{2}} + \frac{x^3}{3} + 2$$

501. $f'(x) = \cos x + \sec^2(x)$, $f\left(\frac{\pi}{4}\right) = 2 + \frac{\sqrt{2}}{2}$

Sol: $f(x) = \sin(x) + \tan(x) + C$. Since $f\left(\frac{\pi}{4}\right) = 2 + \frac{\sqrt{2}}{2}$, then

$$2 + \frac{\sqrt{2}}{2} = f\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) + \tan\left(\frac{\pi}{4}\right) + C = \frac{\sqrt{2}}{2} + 1 + C \Rightarrow C = 1$$

$$\Rightarrow f(x) = \sin(x) + \tan(x) + 1$$